S1: Frequency hopping under partial-band jamming

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Purpose. S1 adds a hopping transmitter and a partial-band noise jammer to the S0 link. This note derives (i) why hopping buys a processing gain, (ii) the closed-form BER used to verify the simulator, (iii) the jammer's optimal band fraction ρ^* and the resulting inverse-linear BER law, (iv) the band fraction that matters for a link *outage*, which S2 builds on, and (v) the jamming margin numbers reported in `results/s1_link_budget.txt`. Partial-band and pulse noise jamming are standard textbook material [1, Sec. 12.6]; the worst-case analysis is written out here because the code uses its constants. Mathematics used: total probability, one-variable optimisation.

1 Processing gain

The data occupies bandwidth W_d; the transmitter hops among N_{ch} channels spread over $W_{ss} \approx N_{\text{ch}} W_d$. A jammer with total power J that wants to cover the whole hopping band must spread J over W_{ss}, so its density is $N_J = J/W_{ss}$. The transmitter's signal at any instant occupies only W_d, where the jammer delivers only $N_J W_d = J\, W_d/W_{ss} = J/N_{\text{ch}}$ of its power. The jammer effectively loses a factor

$$G_p = \frac{W_{ss}}{W_d} \approx N_{\text{ch}}, \qquad G_p\,[\text{dB}] = 10 \log_{10} N_{\text{ch}}.$$

With $N_{\text{ch}} = 100$: $G_p = 20\,\text{dB}$.

Relating E_b/N_J to J/S. With signal power S and bit rate R_b, $E_b = S/R_b$. Taking $W_d \approx R_b$ (a different bandwidth convention changes G_p by a constant and shifts every J/S axis by the same amount),

$$\frac{E_b}{N_J} = \frac{S/R_b}{J/W_{ss}} = \frac{S}{J} \cdot \frac{W_{ss}}{R_b} \approx \frac{S}{J}\, G_p \quad \Longrightarrow \quad \boxed{\frac{E_b}{N_J}\,[\text{dB}] = G_p\,[\text{dB}] - \frac{J}{S}\,[\text{dB}]}.$$

This is the line `ebnj_lin = db_to_linear(gp_db - js_db)` in the code and the reason the figure's x-axis is J/S while the analysis runs on E_b/N_J.

2 Partial-band jamming: the mixture formula

A jammer need not spread over the whole band. It can concentrate J on a fraction $\rho \in (0, 1]$ of the channels, raising its density there to N_J/ρ and leaving the rest clean. The jammer does not know the pseudo-random hopping pattern, so each hop lands in the jammed region independently with probability ρ. By the law of total probability,

$$P_b(\rho) = (1-\rho)\, \text{Q}\!\left(\sqrt{\tfrac{2E_b}{N_0}}\right) + \rho\, \text{Q}\!\left(\sqrt{\tfrac{2E_b}{N_0 + N_J/\rho}}\right). \tag{1}$$

The first term is the thermal-noise-only BER of S0 (clean hop); the second is an S0 link whose noise density is N_0 plus the concentrated jammer density. This is `ber_partial_band_theory()`

and it is what the Monte Carlo (random jammed flag per hop, then the S0 signal model) is checked against, point by point, with the z-score test of the S0 note (`VERIFICATION 1`).

Hop rate. The simulation sends one bit per hop so that successive bits are jammed independently. Eq. (1) is an *average* over hops and holds for any number of bits per hop; what changes with slow hopping is the burstiness of the errors, which matters for coding and is outside this model.

Limit check ($\rho = 1$). Eq. (1) reduces to $\mathrm{Q}\big(\sqrt{2E_b/(N_0+N_J)}\big)$: plain AWGN with the two densities added (`VERIFICATION 2`, agreement to 10^{-12}).

3 The jammer's optimal ρ and the inverse-linear law

From the jammer's point of view ρ is a free parameter and it wants to maximise (1). When the jammer dominates the thermal noise ($N_J/\rho \gg N_0$), drop N_0:

$$P_b(\rho) \approx \rho\, \mathrm{Q}\left(\sqrt{\tfrac{2\rho E_b}{N_J}}\right).$$

Substitute $x = 2\rho\, E_b/N_J$ (so $\rho = x/(2E_b/N_J)$):

$$P_b \approx \frac{x\ \mathrm{Q}(\sqrt{x})}{2\,E_b/N_J} = \frac{g(x)}{2\,E_b/N_J}, \qquad g(x) := x\ \mathrm{Q}(\sqrt{x}).$$

The dependence on E_b/N_J has been pulled out; the jammer just maximises $g(x)$ over $x > 0$. Setting $g'(x) = 0$, with $\mathrm{Q}'(u) = -\varphi(u)$ and $\varphi(u) = e^{-u^2/2}/\sqrt{2\pi}$,

$$g'(x) = \mathrm{Q}(\sqrt{x}) - x\,\varphi(\sqrt{x})\,\frac{1}{2\sqrt{x}} = \mathrm{Q}(\sqrt{x}) - \frac{\sqrt{x}}{2}\,\varphi(\sqrt{x}) = 0.$$

This transcendental equation has one positive root, found numerically:

$$x^* \approx 1.418, \qquad g(x^*) \approx 0.1657, \qquad \mathrm{Q}(\sqrt{x^*}) \approx 0.1169.$$

Numerical check: at $x = 1.418$ both sides of $\mathrm{Q}(\sqrt{x}) = \frac{\sqrt{x}}{2}\varphi(\sqrt{x})$ equal 0.1169.

Two consequences follow.

$$\rho^* = \frac{x^*}{2\,E_b/N_J} \approx \frac{0.709}{E_b/N_J}, \tag{2}$$

$$P_b^{\mathrm{worst}} = \frac{g(x^*)}{2\,E_b/N_J} \approx \frac{0.083}{E_b/N_J}. \tag{3}$$

Eq. (3) is the main result of S1: *against an optimal partial-band jammer the BER decays only as* $1/(E_b/N_J)$, not exponentially as in S0.

Noncoherent BFSK. Frequency hopping is more commonly paired with noncoherent FSK, whose BER in noise of density N is $\frac{1}{2}e^{-E_b/2N}$. The same steps give $P_b(\rho) \approx \frac{\rho}{2}\,e^{-\rho E_b/2N_J}$; setting the derivative to zero gives $\rho^* = 2/(E_b/N_J)$ and $P_b^{\mathrm{worst}} = e^{-1}/(E_b/N_J)$: a different constant, the same inverse-linear law. Coherent BPSK is used here only because it continues the S0 model.

Check of the optimiser. The script finds ρ^* with a bounded one-dimensional search on the exact formula (1). With thermal noise switched off it must return the constants above; `VERIFICATION 3` obtains $\rho^*\, E_b/N_J = 0.7088$ and $P_b^{\mathrm{worst}}\, E_b/N_J = 0.08286$.

The number in the figure. At $J/S = 12\,\text{dB}$ with $G_p = 20\,\text{dB}$, $E_b/N_J = 8\,\text{dB} = 6.31$. Eq. (2) gives $\rho^* = 0.709/6.31 = 0.112$ with thermal noise ignored. Keeping the thermal term N_0 in (1) shifts the optimum upward; the search on the exact formula gives $\rho^* = 0.135$. The curve is flat near its maximum (the BER at $\rho = 0.10$ and at $\rho = 0.16$ is within 3 % of the peak), so the Monte Carlo points confirm the curve; they are too coarse to locate the maximum. The worst-case BER is 1.42×10^{-2} versus 2.70×10^{-3} for a full-band jammer of the same power, a factor of 5.2.

4 The band fraction that causes an outage

S2 declares a link *down* when $P_b > \text{BER}_{\text{out}}$. The jammer then asks a different question: what is the strongest link it can still bring down? By Eq. (3) the worst-case BER reaches BER_{out} when

$$(E_b/N_J)_{\text{out}} = \frac{0.0829}{\text{BER}_{\text{out}}}, \qquad \rho_{\text{out}} = \frac{0.709}{(E_b/N_J)_{\text{out}}} = \frac{\text{BER}_{\text{out}}}{\text{Q}(\sqrt{x^*})} \approx 8.55\,\text{BER}_{\text{out}}. \tag{4}$$

For $\text{BER}_{\text{out}} = 10^{-3}$: $(E_b/N_J)_{\text{out}} = 82.9 = 19.2\,\text{dB}$ and $\rho_{\text{out}} = 0.0086$. This is Eq. (2) evaluated at the outage point; it is much smaller than the 0.135 of the figure because 19.2 dB is a much larger E_b/N_J than 8 dB.

One-channel limit. A jammer facing N_{ch} channels cannot jam less than one of them, so $\rho \geq 1/N_{\text{ch}} = 0.01$. Here $\rho_{\text{out}} = 0.0086$ is just below that limit. The script optimises ρ as a continuous variable, which keeps the link to Eqs. (2)–(3), and reports the one-channel case alongside; the two differ by 0.01 dB (next section).

5 Jamming margin

A link is *up* if $P_b \leq \text{BER}_{\text{out}} = 10^{-3}$. The value is an assumption; the S2 note shows that changing it shifts every threshold by 10 dB per decade and leaves the dependence on network size unchanged. Solve the BER formula for the E_b/N_J at which the criterion is just met, $(E_b/N_J)_{\text{req}}$, then allow an implementation loss L_{sys} (assumed 2 dB; it shifts all margins equally). The jamming margin is the J/S the link can absorb:

$$M_J\,[\text{dB}] = G_p - (E_b/N_J)_{\text{req}} - L_{\text{sys}}.$$

For $N_{\text{ch}} = 100$, $E_b/N_0 = 10\,\text{dB}$, $L_{\text{sys}} = 2\,\text{dB}$:

jammer strategy	optimum ρ	$(E_b/N_J)_{\text{req}}$	M_J
full band, $\rho = 1$	—	9.61 dB	**8.39** dB
optimal ρ (continuous)	0.0095	19.54 dB	**−1.54** dB
optimal $\rho \geq 1/N_{\text{ch}}$	0.0100	19.53 dB	**−1.53** dB

The required 19.54 dB is slightly above the 19.2 dB of Eq. (4) because the thermal noise ($E_b/N_0 = 10\,\text{dB}$) is not negligible inside the jammed channels. The optimal jammer needs 9.9 dB *less* power than a full-band jammer to break the link. The negative margin means that, at $\text{BER}_{\text{out}} = 10^{-3}$ and without coding, 100 hop channels do not protect a BPSK link against a jammer of equal received power.

6 Correspondence with the code

Mathematics	Code
$G_p = 10 \log_{10} N_{\text{ch}}$	`gp_db = linear_to_db(N_CH)`
$E_b/N_J = G_p - J/S$	`ebnj_lin = db_to_linear(gp_db - js_db)`
jammed hop w.p. ρ	`jammed = rng.random(...) < rho`
$\sigma^2 = (N_0 + N_J/\rho)/2$	`sigma_jammed = np.sqrt((n0 + nj / rho) / 2.0)`
Eq. (1)	`ber_partial_band_theory()`
ρ^* on the exact formula	`worst_case_rho()`
$(E_b/N_J)_{\text{req}}$	`required_ebnj_db()` (root of $P_b - \text{BER}_{\text{out}}$)
$1/N_{\text{ch}}$ limit	`RHO_MIN_PHYSICAL`

References

[1] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Prentice Hall, 2001, ch. 12 (Spread-Spectrum Techniques).